

Practice Quiz: Communication (Implementation)

Name: __ Date: __

Part A: Multiple Choice

1. In the multi-agent implementation, one agent's message becomes: A) The other agent's action B) An observation that the receiving agent processes through its inference pipeline C) A direct modification of the other agent's beliefs D) Noise
2. Generalized synchrony is measured using: A) Euclidean distance B) Jensen-Shannon divergence — symmetric, bounded, and interpretable C) The number of messages exchanged D) Physical proximity
3. A synchrony value of 1.0 means: A) Agents disagree completely B) Agents have identical belief distributions over hidden states C) Communication has failed D) One agent is correct
4. Shared priors between agents: A) Prevent communication B) Give agents similar starting beliefs, making convergence faster C) Make agents identical D) Are impossible to implement
5. `generate_message` returns the argmax of beliefs because: A) It's the only option B) The agent communicates its most confident belief — a simple communication protocol C) argmax is always correct D) NumPy requires it
6. Noisy communication channels: A) Speed up convergence B) Slow down or prevent belief convergence by corrupting messages C) Have no effect D) Always improve accuracy
7. Two agents with very different priors: A) Cannot communicate B) Take more rounds to converge because they must overcome initial disagreement C) Converge instantly D) Always deadlock

Part B: Short Answer

1. Two agents start with beliefs $A=[0.9, 0.1]$ and $B=[0.1, 0.9]$. Compute the initial synchrony using JSD. Show your work.

2. Explain how this implementation models an "echo chamber" — agents who already agree reinforcing each other's beliefs. What would the synchrony trajectory look like?
3. Design an extension where agents can choose NOT to communicate (silence as an action). How would this affect the implementation and convergence dynamics?